Learning Outcome 1 - End Assessment

Question 1: Multiple Choice

Circle the letter that corresponds to the correct answer.

1. What is JSX in React?
   a) A JavaScript framework
   b) A JavaScript XML syntax extension
   c) A state management library
   d) A CSS library
2. Which of the following is a lifecycle method in React?
   a) `render`
   b) `componentDidMount`
   c) `useEffect`
   d) `setState`
3. Which hook is used to manage state in functional components?
   a) `useState`
   b) `useEffect`
   c) `useContext`
   d) `useReducer`
4. Which of the following is a valid way to pass props in React?
   a) Using state
   b) Through the constructor
   c) As an attribute in JSX
   d) Through the `render` method
5. What does the Virtual DOM do?
   a) Stores component states
   b) Provides routing in React
   c) Minimizes the number of DOM manipulations
   d) Handles event listeners
6. What is React Router primarily used for?
   a) Managing component state
   b) Navigation between different views
   c) Handling forms
   d) Making API calls
7. Which method unmounts a React component?
   a) `componentWillUnmount`
   b) `componentDidUpdate`
   c) `render`
   d) `shouldComponentUpdate`
8. Which React hook would you use to run side effects?
   a) `useState`
   b) `useEffect`

c) `useReducer`
d) `useMemo`

9. What does Redux help manage in a React application?
a) UI components
b) Global state
c) Event handling
d) DOM updates
10. Which event type in React is responsible for handling input changes?
a) `onClick`
b) `onChange`
c) `onSubmit`
d) `onMouseOver`
11. What is a synthetic event in React?
a) A browser native event
b) A wrapper around browser native events
c) A custom event created by the developer
d) A debugging tool
12. In React, which function is used to handle API calls?
a) `useFetch`
b) `fetch`
c) `Axios`
d) `getRequest`
13. What is the main purpose of the Context API in React?
a) Data fetching
b) State management across components
c) Handling user inputs
d) Event handling
14. What method is used to prevent event bubbling in React?
a) `stopBubbling`
b) `preventPropagation`
c) `stopPropagation`
d) `stopImmediatePropagation`
15. How is a controlled component defined in React?
a) A component that controls event propagation
b) A component whose state is managed by itself
c) A component whose form data is handled by React state
d) A component that cannot re-render
16. What does the `useEffect` hook do?
a) Handles side effects in function components
b) Updates component state
c) Allows conditional rendering
d) Enables lifecycle methods in class components
17. What does the Redux store represent?
a) A place where React components are defined
b) The global state for a React application

c) A collection of React hooks
d) A virtual DOM replica

18. What is the main use of MobX in React applications?
    a) Manage local state
    b) Manage global state
    c) API integrations
    d) Routing
19. What is the purpose of `componentDidUpdate` in React?
    a) To update the state
    b) To perform actions after a component re-renders
    c) To initialize a component
    d) To unmount a component
20. How can you perform routing in React?
    a) Using React hooks
    b) Using Redux
    c) Using React Router
    d) Using event listeners

Question 2: True or False

Indicate whether the statement is **TRUE** or **FALSE**.

1. JSX allows embedding HTML within JavaScript code.
2. The `useState` hook is used for side effects in React components.
3. The Virtual DOM helps improve the performance of React applications.
4. Props are mutable in React components.
5. The `useEffect` hook is equivalent to all three lifecycle methods in class components.
6. Redux is mainly used to manage local component state.
7. React Router helps in managing state transitions.
8. The Context API is a way to avoid passing props through multiple components.
9. A synthetic event in React is a cross-browser wrapper around native events.
10. The `componentWillUnmount` method is used to clean up after a component is removed.
11. React components re-render when their props or state change.
12. In JSX, class names are assigned using the `class` attribute.
13. Lifecycle methods are only available in class components.
14. The `bind()` method is used to bind event handlers in functional components.
15. MobX is a state management tool similar to Redux.
16. A controlled component uses internal state to manage form data.
17. Context API and Redux both solve global state management problems.
18. Nested routing is supported by React Router.
19. Events in React follow the same event delegation as vanilla JavaScript.
20. React Developer Tools is used to debug and inspect components in a browser.

Question 3: Fill in the Blanks

Complete the following sentences using the most appropriate phrase from the given options:

Options:

- Navigation between different components or pages
- JavaScript XML
- State
- JavaScript function that returns JSX
- A component has been rendered to the DOM
- Side effects
- Hooks
- Class components
- Browser's native events
- In-memory representation
- Global state
- Passing data through the component tree without props
- Parent components
- Child components
- `event.stopPropagation()`
- State management libraries
- Clean up resources before a component is removed from the DOM
- Memorize callback functions
- `useParams`
- `useEffect` hook or `async/await`

1. JSX stands for ___________.
2. React Router is used for ___________.
3. The `useState` hook is used to manage ___________ in functional components.
4. The `componentDidMount` lifecycle method is called after ___________.
5. A functional component in React is a ___________.
6. React allows developers to manage state using both ___________ and ___________.
7. Synthetic events are a cross-browser wrapper around ___________.
8. A Virtual DOM in React is an ___________ of the real DOM.
9. The `useEffect` hook is primarily used for handling ___________.
10. Redux helps in managing ___________ in large React applications.
11. The React Context API is used for ___________.
12. Props in React are passed from ___________ to ___________.
13. To prevent an event from bubbling up the DOM, you would use ___________.
14. MobX and Redux are both ___________.
15. The `componentWillUnmount` lifecycle method is used to ___________.
16. The `bind()` method in React is used to bind event handlers to ___________.
17. The `useCallback` hook in React is used to ___________.
18. A controlled component's form data is handled by ___________.
19. URL parameters in React Router are handled using ___________.

20. To handle asynchronous data fetching in React, you can use __________.

One

Customer Learning Outcome 2 – End Assessment

Q1. True or False

Read the following statements about Tailwind CSS carefully and mark **TRUE** if the statement is correct or **FALSE** if it is incorrect.

a) Tailwind CSS does not support grid-based layouts for creating flexible, responsive designs.
b) The mobile-first approach involves designing for smaller screens first and progressively enhancing the layout for larger screens.
c) Tailwind CSS does not allow developers to customize typography or scale font sizes for responsive design.
d) Tailwind CSS provides utility classes to style interactive elements such as buttons, forms, and hover states, ensuring responsiveness.
e) Testing responsiveness and iterating on design are not necessary when using Tailwind CSS because it automatically makes everything responsive.
f) You can extend Tailwind's default theme by modifying the `tailwind.config.js` file.
g) Custom fonts can be added to Tailwind by specifying them in the `fontFamily` section of the theme configuration file.
h) Tailwind CSS plugins can add additional utilities and variants for extended functionality.
i) Conditional styles in Tailwind can be applied by integrating JavaScript logic using libraries like Alpine.js or React's state management.

Q2. Fill in the Blanks

Complete the following sentences using the most appropriate words from the provided choices.

Word Bank:
grid, colors, focus state, flex-initial, tailwind.config.js, hover state, animation, transition, animate-bounce, delay-, mobile-first, media queries*

a) ……………………………… in Tailwind CSS is a feature that allows you to apply styles to an element when a user hovers over it with their cursor.
b) You can add plugins by installing them and then including them in the plugins array of ……………………………….
c) The ……………………………… approach is a design strategy where you prioritize mobile users by designing for smaller screens first and then progressively enhancing for larger screens.
d) Add the ……………………………… utility to make an element bounce up and down, useful for elements like "scroll down" indicators.
e) ……………………………… refers to the ability to apply motion effects to elements, either

using pre-defined utility classes or custom animations.
f) You can customize the primary color by extending the ……………………………… property in `tailwind.config.js`.
g) Use the ……………………………… utilities to control an element's transition delay.
h) Tailwind CSS provides pre-configured ……………………………… through its breakpoint utilities like `sm`, `md`, `lg`, `xl`, and `2xl`.
i) Use ……………………………… to allow a flex item to shrink but not grow, considering its initial size.
j) In Tailwind CSS, you can implement flexible grid layouts using the ……………………………… utility class and specifying the number of columns with `grid-cols-{n}`.

Q3. Match the Columns

Match the items in **Column A** with the correct answers from **Column B**.

Column A	Column B
1. Utility for controlling gutters between grid and flexbox items	A. Auto-cols-min
2. Utility for controlling the size of implicitly-created grid rows	B. shrink-0
3. Utility for controlling the size of implicitly-created grid columns	C. animate-pulse
4. Utility for specifying the rows in a grid layout	D. Gap
5. Prevents a flex item from shrinking	E. grow-0
6. Used to make an element gently fade in and out, useful for things like skeleton loaders	F. Auto-rows-fr
7. Used to add margins to a paragraph	G. grid-rows-5
8. Applies line spacing to text	H. md
9. Example of a breakpoint	I. m-4
10. Applies an extra-large text size	J. leading-normal
	K. text-xl
	L. text-2xl
	M. p-4

Practical Task – ReactJS & Tailwind CSS Development

ABC Company, a software development firm based in Nyanza District, Southern Province – Rwanda, has developed a **ReactJS** application with an unappealing and outdated design. The company seeks to enhance its web development techniques by adopting modern best practices.

As a **ReactJS expert**, you have been hired to redesign the company's application using **ReactJS and Tailwind CSS** to ensure a **modern, responsive, and visually appealing design** across all devices.

Your Tasks:

1. Implement a **clean, consistent, and responsive** UI using Tailwind CSS.
2. Integrate **animations and transitions** to improve the user experience.
3. Handle various **states like hover and focus** for interactive elements.
4. Follow a **mobile-first approach** to ensure seamless usability on different screen sizes.
5. Customize styles to match the company's branding by incorporating:
   - **Custom fonts**
   - **Brand colors**
   - **Unique layout directives**

Your final application should be fully functional and provide an **engaging, intuitive, and modern** user experience.

Q1: True/False

Read the following statements and answer **TRUE** if correct or **FALSE** otherwise.

a) TypeScript requires a separate compiler that is different from JavaScript.
b) TypeScript does not support configuration options for module resolution.
c) Nested routes allow you to create child routes within a parent route.
d) API routes in Next.js cannot handle dynamic segments in the URL.
e) Performing client-side security measures is unnecessary if you are using server-side rendering (SSR).
f) The Link component in Next.js can only navigate to external URLs.
g) Implementing a Content Security Policy (CSP) can help mitigate cross-site scripting (XSS) attacks.

Q2: Multiple Choice

Read the following questions carefully and select the correct answer.

1. Which component is used for client-side navigation in Next.js?
   A) Router
   B) Link
   C) Navigate
   D) Route
2. Dynamic routes in Next.js can be created by using:
   A) [] brackets in the filename
   B) () brackets in the filename
   C) # symbols in the filename
   D) None of the above
3. Which of the following is NOT a request type handled by an API?
   A) GET
   B) POST
   C) PATCH
   D) EXECUTE
4. To secure your Next.js application against cross-origin attacks, you should implement:
   A) CORS
   B) HTTPS
   C) CSP
   D) All of the above

Q3: Fill in the Blanks

Read the following sentences and fill in the missing word(s).

a) The purpose of ___________ is to manage user sessions and authenticate requests on the server side.
b) Using ___________ in Next.js allows you to pass parameters directly in the URL and retrieve them in your component.
c) Programmatic navigation can be achieved using the ___________ method provided by the Next.js router.
d) To create a catch-all route, you would use the syntax ___________ in the filename.
e) Testing your API can be done using tools like ___________ or Postman to ensure the endpoints are functioning correctly.

Q4: Matching Concepts

Match each concept on the left with the correct description on the right.

Concepts	Descriptions
1. Preparation of Environment	A. The process of defining the layout and visual appearance of your application using CSS or a CSS-in-JS library.

Concepts	Descriptions
2. Project Creation	B. The configuration of tools and dependencies, such as Node.js and npm, required to run a Next.js application.
3. Creating Pages and Components	C. A method to enhance web performance by storing and retrieving data from a cache rather than fetching it each time.
4. Implementing SEO	D. The use of meta tags and structured data to improve visibility and ranking in search engines.
5. Styling	E. The fundamental building blocks of a Next.js application, typically stored in the pages directory.
6. Caching Strategies	F. The command npx create-next-app used to initiate a new Next.js project.

Project Task

XYZ Developers is a web application company located in Kigali city. It needs to hire a **Next.js developer** for the company. Assume you are hired as their developer and tasked with preparing a **travel booking platform** using **Next.js and TypeScript**. You are requested to:

- Set up a **Next.js project** with **TypeScript support**.
- Ensure **security** throughout your application.
- Implement **various rendering techniques** such as **server-side rendering (SSR)** and **static site generation (SSG)** to optimize performance.
- Create a **robust API** to handle user requests for searching and booking accommodations, activities, and transportation.
- Use **Next.js routing** to establish seamless navigation between key pages, including **Home, Search Results, and Booking Details**.

Q1: True/False

Read carefully the following statements about Progressive Web Applications (PWA) and answer **TRUE** if the statement is correct and **FALSE** if it is incorrect.

a) Leveraging Progressive Enhancement means starting with a basic version of the application and adding features for more capable browsers.
b) Prioritizing Mobile-First Design involves designing the application primarily for desktop users and then adapting it for mobile devices.
c) Utilizing Performance Optimization Techniques can include minimizing image sizes, optimizing code, and reducing the number of HTTP requests to enhance the loading speed of a React application.
d) Progressive Enhancement ignores the capabilities of modern browsers and focuses solely on older browsers to ensure compatibility.
e) A Mobile-First Design approach typically results in a better user experience on all devices by ensuring that mobile users have a fully functional and optimized interface.
f) Performance optimization techniques have no impact on the responsiveness of a React

application.
g) The web application manifest is a JSON file that provides metadata about the web application.
h) The manifest file can specify the application's name, icons, and start URL.
i) The display property in the manifest controls how the app appears on the user's home screen.
j) Testing and validation of the manifest file can be done using the browser's Developer Tools.
k) The manifest file should be included in the public directory of a React application to be served correctly.
l) A service worker must be registered before it can be installed.
m) A service worker will continue to function even if the user closes the browser.
n) Service workers can only cache static assets; they cannot handle dynamic content.

Q2: Fill in the Blanks

Fill in the blank spaces with appropriate word(s). Select from the given choices in the box:

Choices: `<link>`, `index.js`, `Lazy loading`, `index.css`, `Lighthouse`, `caches.delete()`, `display`, `install`

a) A service worker can be registered in a React application by using the `navigator.serviceWorker.register` method, typically placed in the …………………………… file of the application.
b) You can reference the manifest file by adding a ……………………….. tag in the `index.html` file inside the `public` folder.
c) The …………………………… event is triggered when the service worker is being installed. This is where caching of static assets and other initial setup can be performed.
d) The ………………………………………………… method is used to remove specific cache entries.
e) You can use tools like the ………………………………………… tool in Chrome DevTools or online manifest validators to ensure the manifest file follows best practices and is PWA-compliant.
f) The …………………………………… property is set to `standalone` or `fullscreen` to make it appear like a native app.
g) …………………………………………………… is a performance optimization technique used in web development, where certain resources (like images, videos, scripts, or components) are loaded only when they are needed rather than during the initial page load.

Q3: Matching Concepts

Match each description in **Column A** with the correct **Key Term** in **Column B**.

Column A (Description)	Column B (Key Term)
1. When a new service worker is detected, the existing one is replaced.	A. Service Workers
2. They act as a proxy between the network and the application.	B. Prioritize Mobile-First Design
3. Start design with mobile layouts and scale up to larger screens.	C. CSS Media Queries
4. Serve appropriate image sizes based on device resolution.	D. Lazy Loading
5. Use to apply different styles based on device characteristics.	E. Responsive Images
6. Utilize tools like Lighthouse or Manifest Validator to check your manifest.	F. Name
7. Load resources only when they are needed to improve performance.	G. JSON
8. Property specifies the name of the application in the manifest.	H. Validating the Manifest File with Tools
9. Primary format used to define the manifest file.	I. Updating Service Worker

Project Task

Scenario:

MXN Breads is a bakery company located in **Rwamagana District, Eastern Province, Rwanda**. It needs an **eCommerce website** that will allow customers to buy products online.

Assume the company has **hired you** as a **full-stack developer** and has tasked you with developing a **Progressive Web Application (PWA)** using **React**.

Your tasks:

1. **Set up a React project** and configure it as a **PWA**.
2. **Ensure progressive enhancement** so that the application works well in all browsers.
3. **Prioritize a Mobile-First Design** for a better user experience.
4. **Use performance optimization techniques** such as **lazy loading, code splitting, and caching strategies** to improve loading speed.
5. **Configure the manifest file** to include app metadata such as the name, icons, and display mode.
6. **Implement a caching strategy** using service workers to improve offline support.
7. **Manage the service worker lifecycle** effectively to ensure updates are properly handled.

This structured assessment will test your knowledge of **Progressive Web Applications (PWA), service workers, performance optimization, and React development**. 🚀

Learning Outcome 5 End Assessment

Q1: True/False Questions

a) Environment variables in a React application can be used to store sensitive information such as backend host, FTP host, FTP username, and FTP password.
b) Changing the value of an environment variable requires restarting the development server for the changes to take effect.
c) Vercel supports automatic deployments from a Git repository, allowing for continuous integration.
d) When migrating application files to a deployment platform, it's unnecessary to include configuration files like package.json.
e) React applications can use platform-specific options for environment variable configurations.
f) Testing the deployed application is important to ensure that it works as expected in the production environment.
g) You can directly access environment variables in React without prefixing them with REACT_APP_.
h) It is best practice to keep all environment variables in a single .env file for security purposes.
i) After configuring DNS settings, it is unnecessary to wait for DNS propagation to take effect.
j) You can create separate .env files for different environments (e.g., .env.development, .env.production) in a React application.
k) Migrating application files involves copying only the src folder from your React application.
l) After deploying a React application, you should avoid running any performance audits to ensure the app functions correctly.
m) Verifying SSL settings can be done by accessing the website and checking if the URL starts with "http://" instead of "https://".

Q2: Fill in the Blank Spaces

a) The command …………………………… is used to create a production-ready build of a React application.
b) You can set environment variables in ReactJS by creating a ………………………… file in the root of your project.
c) Add an ……………………………………… to point your domain to the IP address of the server hosting your React application.
d) Variables should be prefixed with ………………………………………… for React to recognize them.
e) Use the command ………………………………… to check if your domain resolves to the correct IP address.
f) The ……………………………… is returned to the browser, which uses it to connect to the

server.
g) The top of the DNS hierarchy is represented by a ………………………………. symbol.

Q3: Matching

1. DNS
2. SSL
3. HTTPS
4. FTP host
5. CNAME Record
6. Top-Level Domain
7. Root Level
8. IP address
9. ISP
10. Domain registrar

Match each term above with one of the following descriptions:
A. It is an extension of HTTP that adds an extra layer of security.
B. It is a type of DNS (Domain Name System) record that maps an alias name to a true or canonical domain name.
C. It is the last part of a domain name, appearing after the final dot. It helps indicate the nature or origin of the domain.
D. The top of the hierarchy, represented by the dot (.) at the end of the domain.
E. It is a system that translates human-readable domain names into machine-readable IP addresses that computers use to identify each other on the network.
F. It is the server or system that provides the File Transfer Protocol service, allowing users to upload, download, and manage files over the internet.
G. It is a company that provides individuals and organizations access to the internet.
H. It is a standard security technology used to establish an encrypted link between a server and a client, typically a web server (website) and a browser.
I. It is the unique identifier for the account.
J. It is a type of DNS (Domain Name System) record that maps a domain name to its corresponding IPv4 address.
K. It is a company or organization that manages the reservation of Internet domain names.
L. It is a unique identifier assigned to devices (such as computers, smartphones, servers) connected to a network that uses the Internet Protocol for communication.

Deployment Scenario

ABC Company is a company located in Kigali City. It hired a React.js developer to develop a customer feedback system. The company wants to hire a ReactJS developer to deploy the application on Vercel. The client has provided a custom domain, **www.abc.com**, and they want the application to be accessible via this domain.

Assume you are hired as a ReactJS developer and tasked with ensuring the application has SSL (HTTPS) enabled for secure access. Your tasks are to:

1. Configure the DNS settings to point the custom domain to the Vercel deployment.
2. Set up SSL to ensure that the site can be accessed via HTTPS.
3. Test and verify that the domain is properly configured and secured.